Parts of Menche et al. (2015) Used in Our Analysis

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Parts used

Our analysis takes ten elements from Menche et al. [1] and its Supplementary Materials [2]: the interactome, the gene-overlap measures, the localization measures, the separation measure and the published code. The results sections below report the notebook outputs as printed, with the method used for each.

What we used	Where in Menche et al.	Where in our work
Human interactome: 141,296 physical interactions between 13,460 proteins, keyed by Entrez Gene ID. The paper does not explicitly specify that the reported Gene IDs correspond to Entrez Gene IDs. This was inferred based on the fact that all of the reported IDs can be found in the Entrez Gene database.	Data S1 [2]; main text, Materials and Methods [1]	`data_process_entrez.ipynb`
The seven interaction types (regulatory, binary, literature, metabolic, complexes, kinase, signaling)	SM section 1.1.1 [2]	`data_source` column of the interactome files
Removal of self-interactions	Source code, `localization.py` [2]	`data_process_entrez.ipynb`

What we used	Where in Menche et al.	Where in our work
Overlap coefficient C and Jaccard index J for two gene sets	SM section 3.1, Eqs. S13 and S14 [2]	`disease_gene_overlap.ipynb`, gene overlap
Hypergeometric model and expected overlap of two gene sets	SM section 3.1, Eq. S15 [2]	`disease_gene_overlap.ipynb`, gene overlap
Module size S: the largest connected component of a disease's genes	SM section 2.1 [2];	`disease_gene_overlap.ipynb`, localization
Shortest distance d_s: from each disease gene to the nearest other gene of the same disease	SM section 2.1 [2]	`disease_gene_overlap.ipynb`, localization
z-score of S against random gene sets, with the 1.6 threshold	SM section 2.1, Eq. S8 [2]	`disease_gene_overlap.ipynb`, localization
Network separation s_AB, with shared genes counted as distance 0	SM section 2.2, Eq. S9 [2]; main text, Fig. 3A–C [1]	`disease_gene_overlap.ipynb`, separation
Full randomization model and z-score for s_AB	SM section 2.2.1, Eq. S10 [2]	`disease_gene_overlap.ipynb`, separation

The definitions in our notebook follow the published programs `localization.py` and `separation.py` [2]. Our saved gene lists use the same format as their example files `MS.txt` and `PD.txt`: one gene per line.

Definitions and formulas

Every term and formula used in this report is collected here. Formulas are written in plain text, as implemented in our notebooks, with their source in Menche et al.

Notation

Symbol	Meaning
A, B	The gene sets of two diseases, restricted to genes on the network
n_A, n_B	Number of genes in A and in B
n_AB	Number of genes that are in both A and B (shared genes)
N	Number of genes in the mapped network (13270)
d(a, b)	Shortest distance between genes a and b: the smallest number of interactions on a path between them
⟨x⟩	Mean of x
mean_rand, sd_rand	Mean and standard deviation of a quantity over the random gene sets
choose(n, k)	Number of ways to pick k items out of n (binomial coefficient)

Data terms

Term	Definition
Interactome	The network of physical protein interactions from Data S1 [2]. Genes are nodes and interactions are edges
Data source	The evidence type(s) of an interaction: regulatory, binary, literature, metabolic, complexes, kinase or signaling
Self-interaction	An interaction whose two gene IDs are the same
Duplicate pair	The same two genes listed twice; (A, B) and (B, A) count as one interaction

Term	Definition
Unmapped gene	A gene whose Entrez ID has no symbol in the HGNC complete set [3]
Unmapped interaction	An interaction where at least one of the two genes is unmapped
Mapped network	The interactome after removing self-interactions, duplicate pairs and unmapped interactions
globalScore	The overall association score of a gene with a disease in Open Targets, from 0 to 1
Disease gene set	The genes of a disease with globalScore ⩾ 0.4 that are nodes of the mapped network

Gene overlap

Quantity	Formula	Meaning	Source
Jaccard index J	J = n_AB / (n_A + n_B − n_AB)	Shared genes as a fraction of all genes in either set. 0 = nothing in common, 1 = identical	SM section 3.1, Eqs. S13 and S14 [2]
Overlap coefficient C	C = n_AB / min(n_A, n_B)	Shared genes as a fraction of the smaller set	SM section 3.1, Eqs. S13 and S14 [2]
Expected overlap	expected = n_A × n_B / N	Shared genes expected if both sets were drawn at random from the network	SM section 3.1, Eq. S15 [2]
Fold	fold = n_AB / expected	Above 1 means more shared genes than expected by chance	Our addition

Quantity	Formula	Meaning	Source
Hypergeometric p-value	p = sum over i from n_AB to min(n_A, n_B) of choose(n_A, i) × choose(N − n_A, n_B − i) / choose(N, n_B)	Probability of at least n_AB shared genes by chance	SM section 3.1 [2]

Localization

Quantity	Formula	Meaning	Source
Module size S (LCC size)	S = number of genes in the largest connected component of the subgraph formed by the genes of A	The largest group of disease genes linked to each other directly or through other disease genes	SM section 2.1 [2]
LCC fraction	LCC fraction = S / n_A	Share of the disease genes that belong to the module	Our addition
Shortest distance d_s	d_s = mean over genes a in A of the smallest d(a, b) among the other genes b in A	How close each disease gene is to the nearest other gene of the same disease. Genes with no path to any other gene of the set are left out of the mean	SM section 2.1 [2]

Separation

Quantity	Formula	Meaning	Source
d_AA, d_BB	d_AA = d_s of A; d_BB = d_s of B	Mean shortest distance within each disease	SM section 2.2 [2]

Quantity	Formula	Meaning	Source
d_AB	d_AB = (sum over a in A of the smallest d(a, b) with b in B + sum over b in B of the smallest d(a, b) with a in A) / (n_A + n_B)	Mean shortest distance between the two diseases. A shared gene has distance 0	SM section 2.2 [2]
Separation s_AB	s_AB = d_AB − (d_AA + d_BB) / 2	s_AB < 0: the two disease modules overlap. s_AB > 0: they are separated	SM section 2.2, Eq. S9 [2]; main text, Fig. 3A–C [1]

Statistical significance

Quantity	Formula	Meaning	Source
Random gene set	Same number of genes as the real set, drawn uniformly from all N network genes; 10000 repeats, random seed 42	The full randomization model. For s_AB, both sets are replaced by random sets	SM section 2.2.1 [2]
z-score	z = (observed − mean_rand) / sd_rand	How many standard deviations the observed value lies from the random mean. For S, z > 1.6 marks a disease module. For d_s and s_AB, a negative z means closer than random	SM sections 2.1 and 2.2.1, Eqs. S8 and S10 [2]
Empirical p-value	p = (r + 1) / (10000 + 1)	r is the number of random values at least as extreme as the observed one: at least as large for S, at least as small for d_s and s_AB. One-sided	Our addition

Data processing

The raw interactome of Menche et al. [2] is converted into a symbol-keyed network of 137863 interactions between 13270 genes, and each disease is reduced to the genes that lie on that network. The interactome steps are in `data_process_entrez.ipynb`; the disease-gene steps are in `disease_gene_overlap.ipynb`.

Input data

Input	File	Content	Source
Human interactome	`DataS1_interactome.tsv`	141296 interactions between 13460 genes; columns: two Entrez Gene IDs and the evidence type(s)	Data S1 [2]
Gene nomenclature	`hgnc_complete_set.txt`	45111 approved HGNC records; columns used: `entrez_id`, `symbol`	HGNC complete set [3], downloaded 25 September 2026
Disease–gene associations	`Breast Cancer.tsv`, `hypertensive disorder-MONDO_0005044.tsv`, `systolic blood pressure-EFO_0006335.tsv`	One row per gene; columns used: `symbol`, `globalScore`	Open Targets Platform association exports

Interactome

1. **Read.** The interactome file is read as tab-separated text. Its header lines, which start with `#`, are skipped. The three columns are named `gene_ID_1`, `gene_ID_2` and `data_source`.
2. **Build the identifier map.** The `entrez_id` and `symbol` columns of the HGNC complete set [3] are taken, rows without an Entrez ID are dropped, and the IDs are converted to integers. This gives 44432 Entrez ID → symbol pairs. Two Entrez IDs occur twice in the HGNC file; the later row is used.

3. **Map to symbols.** Both gene columns are mapped through this table, giving `symbol_1` and `symbol_2`. Of the 13460 genes, 13334 (99.1%) receive a symbol. The 126 without one are saved and checked against NCBI Gene.
4. **Remove self-interactions.** Rows where both gene IDs are the same are removed, as in the Menche code [2]. This leaves 138427 of 141296 rows.
5. **Remove duplicate pairs.** (A, B) and (B, A) are treated as one interaction. No duplicates are found, so 138427 rows remain. This table is saved as `interactome_all.csv`.
6. **Keep fully mapped interactions.** Rows where either gene has no symbol are removed (564 rows). The remaining 137863 interactions between 13270 genes are saved as `interactome_both_symbols.csv`, the network used in every later step.

Disease gene sets

1. **Read.** Each disease file is read as tab-separated text.
2. **Filter by score.** Genes with `globalScore` ⩾ 0.4 are kept.
3. **Restrict to the network.** Of those, only genes that are nodes of the mapped network are kept, matched by gene symbol. This gives 601 genes for BC, 308 for HTN and 492 for SBP.
4. **Save.** Each set is written to `gene_lists/` with one symbol per line, the format of the Menche example files [2].

Note on the cutoff. The `globalScore` cutoff of 0.4 is arbitrary. Open Targets does not define a threshold for this score, and we have no documented reason for choosing this number. We may need a better justification for a specific value.

Results: interactome processing

These are the outputs of `data_process_entrez.ipynb`, with the method used for each.

Identifier mapping

Method. The interactome file (Data S1 [2]) is read with its comment lines skipped. The `entrez_id` and `symbol` columns of the HGNC complete set [3] give an Entrez ID → symbol map; rows with a missing value are dropped. Both gene columns of the interactome are mapped through it.

Quantity	Value
Entrez IDs in the HGNC map	44432
Genes in interactome	13460
Genes with a symbol	13334 (99.1%)

Unmapped genes and interactions

Method. An interaction is counted as unmapped when at least one of its two genes has no symbol. The count is made on the raw file and again after self-interactions are removed.

Quantity	Raw file	After removing self-interactions
Interactions	141296	138427
Unmapped interactions	575 (0.41%)	564 (0.41%)
One gene unmapped	562	562
Both genes unmapped	13	2
Genes	13460	13397
Unmapped genes	126 (0.94%)	124 (0.93%)

The 126 unmapped Entrez IDs are saved to `unmapped_gene_ids.txt`, and their interaction counts to `unmapped_genes.csv`. Each ID is then looked up in NCBI Gene for its record status:

NCBI status	IDs
replaced	74
discontinued	31
live	21

Network cleaning

Method. Rows where both gene IDs are identical are removed, as in the Menche code [2]. Duplicate pairs are then removed, treating (A, B) and (B, A) as the same interaction. Finally, only interactions where both genes have a symbol are kept.

Step	Rows before	Rows after
Remove self-interactions	141296	138427 (−2.03%)
Remove duplicate gene pairs	138427	138427
Keep interactions with both symbols	138427	137863

The mapped network has 137863 interactions and 13270 genes. The 564 rows with a missing symbol are saved to `missing_symbols.xlsx`.

Sanity check

Method. Five known interacting pairs are searched for in the mapped network.

Pair	Result
TP53 – MDM2	found
BRCA1 – BARD1	found
BRCA2 – PALB2	found
ESR1 – NCOA3	found
CDK4 – CCND1	found

Results: disease genes and gene overlap

These are the outputs of the first half of `disease_gene_overlap.ipynb`. The diseases are breast cancer (BC), hypertensive disorder (HTN) and systolic blood pressure (SBP).

Network

Method. The network is built from `interactome_both_symbols.csv`, with gene symbols as nodes.

Quantity	Value
Genes (nodes)	13270
Interactions (edges)	137863

Disease gene sets

Method. For each disease file, genes with `globalScore` ⩾ 0.4 are kept. Of those, only the genes that are nodes in the network are kept. The percentage is on-network genes divided by genes with score ⩾ 0.4.

Disease	All genes in file	Genes with score ⩾ 0.4	On the network	Percentage
BC	17064	629	601	95.548490
HTN	10012	377	308	81.697613
SBP	4884	635	492	77.480315

Gene overlap between diseases

Method. For each pair of gene sets, the measures defined in "Definitions and formulas", under "Gene overlap", are computed, with N = 13270 network genes.

Measure	BC – HTN	BC – SBP	HTN – SBP
Size A	601	601	308
Size B	308	492	492
Shared	46	64	176
Jaccard	0.053302	0.062196	0.282051
Overlap coef.	0.149351	0.130081	0.571429
Expected	13.949359	22.282743	11.419442
Fold	3.297642	2.872178	15.412311
p-value	6.809469e-13	1.446216e-14	4.976575e-179

Shared genes

Method. The shared genes are the set intersection of the on-network gene lists, sorted by symbol.

BC & HTN (46 genes): ABL1, ASXL1, CACNA1D, CCDC170, CDKAL1, CDKN1B, CUX1, EBF1, FES, FTO, HCK, HMGCR, INHBB, INPP4B, JAZF1, KDR, KIT, LEF1, LSP1, MECOM, NFATC2, NRIP1, NT5C2, PDGFRA, PDGFRB, PIK3CG, PRDM16, PTGS2, PTPN11, PTPRT, REEP3, SETBP1, SH2B3, SLC6A2, SOS1, STN1, TBX3, TCF7L2, TERT, TNS1, WT1, YWHAE, ZFP36L1, ZFPM2, ZNF521, ZNRF3

BC & SBP (64 genes): ADCY3, ARHGEF12, BCL2, CACNA1D, CCDC6, CDH11, CDKAL1, CDKN1B, CREB3L1, CTNNB1, EBF1, EMID1, ERG, ESR1, EXO1, FES, FGFR2, FLI1, FOXO3, FTO, GATA2, GNAS, HLF, HMGA2, HNF4G, INPP4B, JAZF1, KDM5A, LATS2, LEF1, LSP1, MECOM, MITF, MLLT10, MSH6, NFATC2, NRG1, NRIP1, NT5C2, OSR1, PDGFRA, PIK3CG, PIK3R1, PPARG, PRDM16, PREX2, PTEN, PTPN11, RAC1, SETBP1, SH2B3, STN1, TAL1, TBX3, TCF7L2, TERT, TNS1, TP63, TTC28, WT1, ZBTB38, ZFHX3, ZFP36L1, ZFPM2

HTN & SBP (176 genes): ABCC8, ACE, ADCY9, ADH1B, ADM, ADRB1, AGT, AKR1B10, APOE, ARHGAP24, ATG7, ATP2B1, BMAL1, BMP4, BMPR1B, C1GALT1, CACNA1D, CACNB2, CDC16, CDH13, CDKAL1, CDKN1B, CEP68, CHD6, CHMP1A, CLIC4, CMIP, CPS1, CRIM1, DBH, DPYSL2, DUSP16, EBF1, EBF2, EDNRA, EFEMP1, ENPEP, ERAP1, FAM193A, FBN2, FES, FGF5, FOSL2, FOXC1, FTO, FURIN, GATA4, GATA5, GLIS3, GOSR2, GTF2I, GUCY1A1, HAAO, HDAC7, HECTD4, HIPK2, HTR4, INPP4B, INPP5A, INSR, IRAK1BP1, JAG1, JAZF1, JPH2, KAT2B, KCNJ11, KCNK3, KDM2B, KDM4B, LCORL, LEF1, LFNG, LPA, LSP1, MAPK4, MECOM, MEF2C, MFAP5, MME, MPPED2, MRAP, MRAS, MST1, NAB1, NCALD, NEDD4L, NFATC2, NGF, NPNT, NPPB, NPR1, NPR3, NR3C2, NRIP1, NT5C2, PALM2AKAP2, PCCB, PCDH18, PDE10A, PDE1A, PDE3A, PDGFC, PDGFRA, PHC3, PHETA1, PIK3CG, PKD1, PKN2, PLCE1, PLEKHA7, PLPP3, PODXL, POM121C, PPL, PRDM16, PREX1, PRKAG2, PRKCE, PTPN11, PTPRJ, RGS6, RORA, RP1L1, RUNX3, RXFP2, RYK, SBF2, SEPTIN9, SERPINI1, SETBP1, SH2B3, SH3PXD2A, SHH, SHROOM3, SIPA1L1, SLC2A4, SLC4A10, SLC4A7, SLC7A1, SLC8A1, SMOC1, SNX31, SORCS3, SOS2, SOX6, SOX7, STN1, SVEP1, SWAP70, TBX2, TBX3, TBX5, TCF21, TCF7L2, TERT, TGFB2, THADA, TNS1, TNS2, TP53INP1, TRHR, UBASH3B, UHRF1, UMOD, USP38, VDR, VTN, WNT2B, WT1, YES1, ZBTB10, ZC3HC1, ZFP36L1, ZFPM2, ZNF423, ZNF652

Shared by all three (28 genes): CACNA1D, CDKAL1, CDKN1B, EBF1, FES, FTO, INPP4B, JAZF1, LEF1, LSP1, MECOM, NFATC2, NRIP1, NT5C2, PDGFRA, PIK3CG, PRDM16, PTPN11, SETBP1, SH2B3, STN1, TBX3, TCF7L2, TERT, TNS1, WT1, ZFP36L1, ZFPM2

Overlap at other cutoffs

Method. The same overlap calculation is repeated with `globalScore` cutoffs of 0.3, 0.4 and 0.5.

Cutoff	Pair	Size A	Size B	Shared	Fold	p-value
0.3	BC – HTN	940	726	175	3.402863	6.701614e-51
0.3	BC – SBP	940	1022	149	2.058157	2.298013e-18
0.3	HTN – SBP	726	1022	384	6.867752	4.274591e-253
0.4	BC – HTN	601	308	46	3.297642	6.809469e-13
0.4	BC – SBP	601	492	64	2.872178	1.446216e-14
0.4	HTN – SBP	308	492	176	15.412311	4.976575e-179
0.5	BC – HTN	317	113	7	2.593172	1.864043e-02
0.5	BC – SBP	317	127	15	4.944236	3.534437e-07
0.5	HTN – SBP	113	127	49	45.309038	5.524326e-72

Results: network localization and separation

These are the outputs of the second half of `disease_gene_overlap.ipynb`, run on the gene sets at cutoff 0.4.

Distances and random gene sets

Method. The shortest distance between every pair of genes is computed once by breadth-first search from each gene, and stored in a matrix. Every real or random gene set then looks its distances up in this matrix.

Quantity	Value
Distance matrix	13270 x 13270 genes, 176 MB
Longest distance	13 steps

Method. Each observed value is compared with 10000 random gene sets, using the random model, z-score and empirical p-value defined in "Definitions and formulas", under "Statistical significance".

Localization

Method. The module size S, the LCC fraction and the shortest distance d_s defined in "Definitions and formulas", under "Localization", are computed for each disease.

Measure	BC	HTN	SBP
Genes on network	601	308	492
LCC size S	521	87	198
LCC fraction	0.866889	0.282468	0.402439
Random LCC (mean)	156.6544	32.5118	102.9796
LCC z-score	12.773279	3.283323	3.441561
LCC p-value	0.0001	0.0014	0.0002
d_s	1.133111	1.616883	1.563008
Random d_s (mean)	1.688188	1.847909	1.736259
d_s z-score	−15.012544	−4.620540	−4.253054
d_s p-value	0.0001	0.0001	0.0001

Separation

Method. The separation s_AB and its components d_AA, d_BB and d_AB, defined in "Definitions and formulas", under "Separation", are computed for each pair of diseases.

Measure	BC – HTN	BC – SBP	HTN – SBP
d_AA	1.133111	1.133111	1.616883
d_BB	1.616883	1.563008	1.563008
d_AB	1.321232	1.228728	0.945000
s_AB	−0.053765	−0.119332	−0.644946
Random s_AB (mean)	−0.029115	−0.068157	−0.039860
z-score	−0.859962	−2.009801	−20.657195

Measure	BC – HTN	BC – SBP	HTN – SBP
p-value (closer than random)	0.194181	0.023598	0.000100

Saved files

All files are in `outputs/cutoff_0.4/`.

File	Content
`gene_lists/BC.txt`	601 genes
`gene_lists/HTN.txt`	308 genes
`gene_lists/SBP.txt`	492 genes
`gene_overlap.csv`	gene overlap table
`localization_uniform.csv`	localization table
`separation_uniform.csv`	separation table

References

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2. Menche J, Sharma A, Kitsak M, Ghiassian SD, Vidal M, Loscalzo J, Barabási A-L. Supplementary Materials for [1]: supplementary text (SM sections 1–11), Databases S1–S4, and source code (`localization.py`, `separation.py`). *Science*. 2015. Published with [1] at the same DOI.
3. HGNC Database, HUGO Gene Nomenclature Committee (HGNC), Department of Haematology, Long Road, Cambridge CB2 0PT, United Kingdom. www.genenames.org. Complete HGNC dataset, file `hgnc_complete_set.txt` (tab-separated), from the "Complete dataset download links" section of Statistics & download files; direct file link. Retrieved 25 September 2026. RRID:SCR_002827.